

CO2 — ASSESSMENT TOOL 1

Scenario-Based Requirement Analysis

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AI-Powered Smart Textile Manufacturing Quality Inspection System

Code of Conduct

I, Saruhaasini A (192511400), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

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19. AI-Powered Smart Textile Manufacturing Quality Inspection System

Industry Context

The textile industry demands consistent product quality while maintaining high production speeds. Manual inspection methods often fail to identify fabric defects accurately and consistently. AI, Computer Vision, Industrial IoT, and cloud analytics enable automated quality inspection and production optimization.

Problem Statement

Textile manufacturers experience production losses due to undetected defects, inconsistent quality inspection, and delayed corrective actions. Manual inspection is time-consuming and susceptible to human error. An AI-powered quality inspection platform should analyze fabric images using computer vision, detect defects automatically, classify defect types, monitor production quality, and generate quality analytics dashboards. The system should improve manufacturing efficiency and reduce product rejection rates.

1. Scenario Understanding and Problem Analysis

1.1 Business Domain

The scenario is situated in the textile manufacturing domain, specifically the fabric-production and quality-assurance stage of the supply chain. This domain sits at the intersection of industrial manufacturing, computer vision / artificial intelligence, and Industrial IoT, and is characterized by continuous, high-speed production lines where product quality must be verified without interrupting throughput.

1.2 Stakeholders

Stakeholder	Role in the Scenario
Textile Plant Owner / Top Management	Sponsors quality-improvement initiatives and reviews strategic quality KPIs.
Production / Plant Manager	Oversees daily production flow and acts on quality alerts and reports.
QC Inspectors	Verify AI-flagged defects and provide domain expertise for edge cases.
Machine Operators	Operate looms/weaving machines and respond to real-time quality alerts.
IT / Software Development Team	Design, build, and maintain the AI inspection platform.
Data Scientists / AI Engineers	Train and continuously improve defect-detection models.
Customers / Buyers	Set quality expectations and rely on inspection data for trust and compliance.
Regulatory / Compliance Bodies	Define quality and safety standards the system's outputs must satisfy.

1.3 Existing Challenges

- Manual inspection speed is capped by human visual processing, forcing a trade-off between production speed and inspection thoroughness.
- Detection accuracy is inconsistent, varying by inspector, shift, and fatigue level, resulting in poor repeatability of quality standards.

- Defect data is captured on paper or isolated spreadsheets, making trend analysis and root-cause investigation difficult.
- Feedback to the production line is delayed until end-of-shift or end-of-batch, allowing defective fabric to accumulate before correction.
- Scaling inspection capacity requires hiring and training more inspectors — a slow, linear, and costly response to rising production volume.

1.4 System Objectives

1. Detect textile defects automatically using AI-based computer vision.
2. Classify fabric quality and defect types with minimal human intervention.
3. Continuously monitor production-line quality in real time.
4. Reduce the number of defective products reaching packaging and dispatch.
5. Automatically generate quality-inspection reports and analytics dashboards.
6. Improve manufacturing efficiency and reduce total inspection time.
7. Support predictive quality control through historical trend analysis.

2. Actors and Use Cases

2.1 Actor Identification

The system involves four primary human actors who interact with it directly during daily operation, and two secondary actors — one human, one external system — who interact with it less frequently but are essential to the system's lifecycle and integration.

Actor	Type	Description
Machine Operator	Primary	Operates production-line machinery; initiates/monitors image capture; responds to real-time alerts by pausing or adjusting equipment.
QC Inspector	Primary	Reviews AI-flagged defects, confirms or overrides classifications, and performs final quality sign-off.
Production / Plant Manager	Primary	Monitors the live quality dashboard, reviews reports, and receives threshold-based alerts.
System Administrator	Primary	Configures camera/lighting settings, manages user access control, and maintains system health.
Data Scientist / AI Engineer	Secondary	Retrains and validates the AI defect-detection model using newly labelled data.
ERP / MES System	Secondary (external system)	Receives synchronized quality data from the platform via API integration.

2.2 Use Case List

Use Case ID	Use Case Name	Actor(s)	Description
UC-01	Capture Fabric Images	Machine Operator	Continuously capture fabric images from the production-line camera.
UC-02	Configure Camera & Lighting	System Administrator	Adjust camera resolution and lighting parameters per production line.
UC-03	Detect Fabric Defects (AI)	System (automated)	Apply AI/computer-vision models to identify visual defects in

Use Case ID	Use Case Name	Actor(s)	Description
			captured images.
UC-04	Classify & Score Defects	System (automated)	Categorize each detected defect and assign a severity score.
UC-05	Confirm / Override Classification	QC Inspector	Manually validate or correct an AI-generated defect classification.
UC-06	View Live Quality Dashboard	Production Manager, QC Inspector	View real-time defect rates and quality metrics by production line.
UC-07	Receive Real-Time Alerts	Machine Operator, Production Manager	Receive threshold-based alerts when defect rates exceed configured limits.
UC-08	Generate Quality Reports	System (automated)	Automatically compile shift-wise and daily quality-inspection reports.
UC-09	Export Reports	Production Manager	Export generated reports in PDF/Excel format for audits and sharing.
UC-10	Manage User Access	System Administrator	Create, modify, or revoke role-based access for platform users.
UC-11	Retrain AI Model	Data Scientist	Retrain the defect-detection model using newly labelled defect data.
UC-12	Sync with ERP / MES	System Administrator, ERP/MES System	Synchronize quality data with the organization's existing ERP/MES system.

2.3 Use Case Diagram

The diagram below represents the interactions between all identified actors and the system's use cases, following UML conventions. The Machine Operator and QC Inspector interact primarily with the detection and verification workflow; the Production Manager and System Administrator interact with monitoring, reporting, and configuration functions; and the two secondary actors interact with model retraining and external system integration respectively. Sequential <<include>> dependencies link image capture to detection to classification, while confirmation/override is modelled as an <<extend>> of the classification use case.

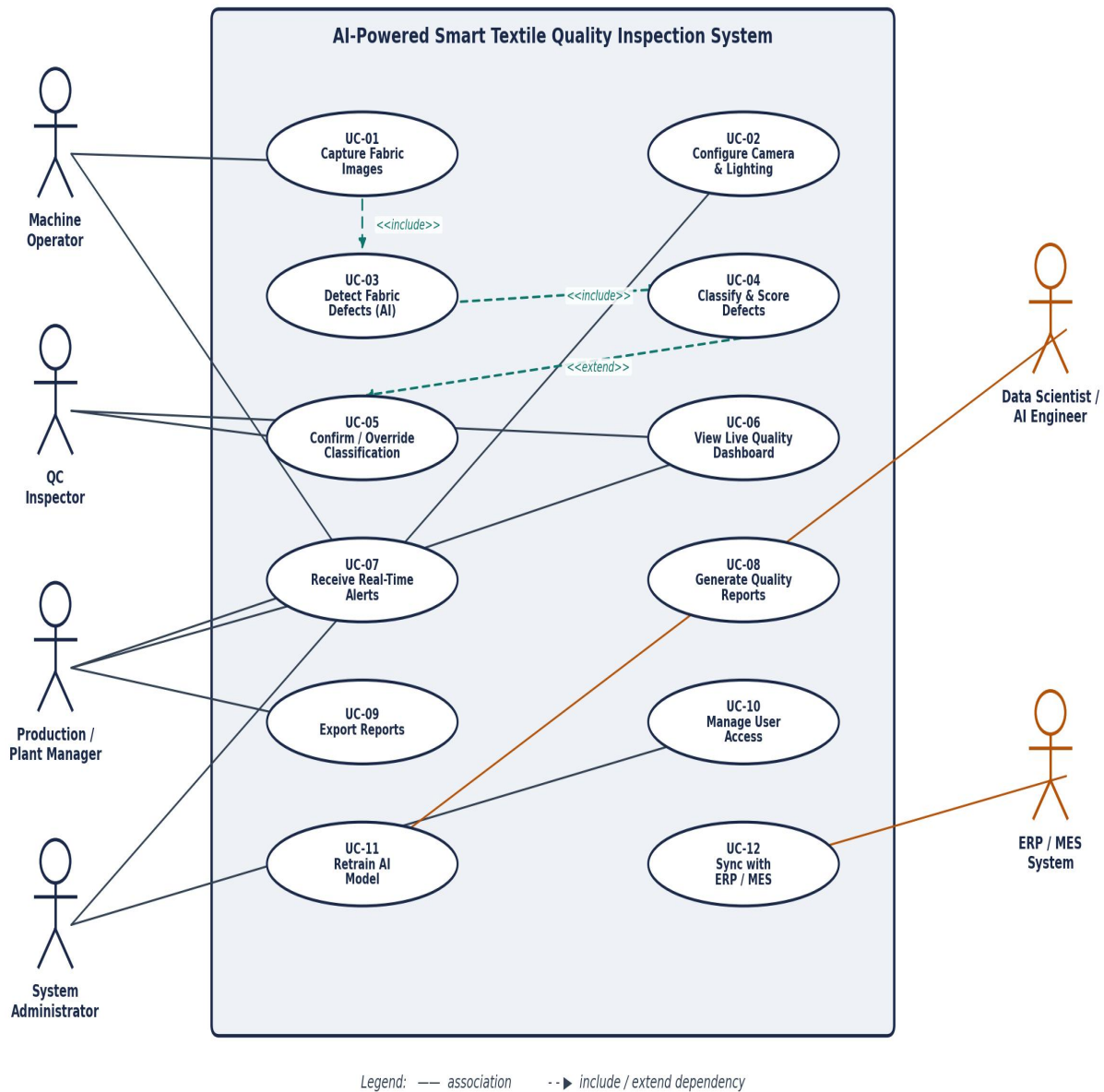


Figure 2.1 — UML Use Case Diagram for the AI-Powered Smart Textile Quality Inspection System.

3. Software Requirement Specification (SRS)

3.1 Introduction

This Software Requirement Specification (SRS) defines the functional and non-functional requirements for the AI-Powered Smart Textile Manufacturing Quality Inspection System. It is intended to guide the design, development, and validation of a platform that automates fabric-defect detection using artificial intelligence, computer vision, and Industrial IoT, replacing the manual inspection process currently used by textile manufacturers.

3.2 Problem Description

Textile manufacturers rely on manual visual inspection to identify fabric defects, a process that is slow, inconsistent across inspectors and shifts, and prone to fatigue-driven error. Because feedback to the production line is delayed until the end of a shift or batch, defective fabric can continue to be produced for extended periods before a fault is corrected. Additionally, defect data is typically recorded on paper or in isolated spreadsheets, leaving manufacturers without the structured data needed for trend analysis or predictive quality control. The proposed system addresses these gaps by automating detection, classification, and reporting through AI-driven analysis of continuously captured fabric images.

3.3 Scope of the System

The system covers real-time image acquisition from production-line cameras, AI/computer-vision-based defect detection and classification, a live quality-monitoring dashboard, automated alerting, automated reporting, and integration with existing ERP/MES systems. It is intended for woven and knitted fabric production lines in small-to-large textile manufacturing units. Physical redesign of looms or weaving machinery, raw-material (yarn/fibre) quality testing, and dyeing-chemistry process control fall outside the scope of this software solution.

3.4 Functional Requirements

ID	Requirement
FR-01	The system shall automatically capture fabric images from production-line cameras continuously.
FR-02	The system shall pre-process captured images (noise reduction, normalization) before analysis.
FR-03	The system shall allow administrators to configure camera and lighting parameters per production line.
FR-04	The system shall automatically detect defects (holes, stains, weaving faults, colour variation) using AI/computer-vision models.

ID	Requirement
FR-05	The system shall classify each detected defect into a predefined defect category.
FR-06	The system shall assign a severity score to each detected defect.
FR-07	The system shall allow QC inspectors to confirm or override AI-generated defect classifications.
FR-08	The system shall display a real-time quality dashboard showing defect rates by production line.
FR-09	The system shall send real-time alerts to operators/managers when defect rates exceed configured thresholds.
FR-10	The system shall automatically generate shift-wise and daily quality-inspection reports.
FR-11	The system shall allow authorized users to export reports in PDF/Excel format.
FR-12	The system shall provide role-based access control for all user categories.
FR-13	The system shall support integration with existing ERP/MES systems via defined APIs.
FR-14	The system shall allow retraining of the AI defect-detection model with newly labelled data.

3.5 Non-Functional Requirements

Category	Requirement
Performance	Defect-detection inference shall complete within sub-second latency per frame to match production-line speed.
Security	All data shall be encrypted in transit and at rest; access shall be governed by role-based permissions.
Reliability / Availability	The system shall maintain at least 99.5% uptime during production hours, with edge-only fallback during cloud outages.
Usability	Dashboards and alerts shall be understandable to shop-floor staff without requiring data-science expertise.
Scalability	The platform shall scale horizontally to support additional production lines or factories without redesign.
Maintainability	AI models shall be retrainable and redeployable without taking the full system offline.
Portability	The quality dashboard shall be accessible via standard web browsers

Category	Requirement
	across desktop and tablet devices.
Interoperability	The system shall expose standard REST/JSON APIs for integration with ERP/MES and other enterprise systems.

3.6 Use Case Descriptions

Detailed flows are provided below for five representative use cases spanning the core detection pipeline, human verification, alerting, and administration.

UC ID / Name	Actor	Preconditions	Postconditions	Main Flow
UC-01 Capture Fabric Images	Machine Operator	Camera is powered on and connected; production line is active.	System is continuously receiving fabric image frames.	1. Operator starts the production line. 2. Camera detects motion/fabric flow. 3. System captures frames at configured rate. 4. Frames are queued for pre-processing.
UC-03 Detect Fabric Defects (AI)	System (automated)	A pre-processed image is available for analysis.	Image is marked as defective or passed, with defect metadata recorded if applicable.	1. AI engine receives pre-processed image. 2. Model performs inference. 3. System flags defects with confidence score. 4. Result is passed to classification module.
UC-05 Confirm / Override Classification	QC Inspector	An image has been flagged as defective by the AI engine.	Final, verified classification is recorded in the quality database.	1. Inspector opens flagged image in inspection UI. 2. Inspector reviews AI classification and severity. 3. Inspector confirms or edits the classification. 4. System records final verified result.
UC-07 Receive Real-Time	Machine Operator, Production	Defect-rate monitoring is active for the	Relevant personnel are notified and	1. System continuously computes rolling defect rate. 2. Rate exceeds

UC ID / Name	Actor	Preconditions	Postconditions	Main Flow
Alerts	Manager	production line.	corrective action can begin.	configured threshold. 3. System pushes alert via UI/SMS/app. 4. Operator/manager acknowledges and acts.
UC-10 Manage User Access	System Administrator	Administrator is authenticated with sufficient privileges.	User accounts and role assignments are updated accordingly.	1. Administrator opens user-management panel. 2. Administrator creates/modifies/revokes a user account or role. 3. System validates and applies the change. 4. System logs the access-control change for audit.

3.7 Data Requirements

- Fabric image data — raw and pre-processed frames captured from production-line cameras.
- Defect metadata — defect type, severity score, bounding-box location, and timestamp for each detection.
- User account data — credentials, roles, and permission levels for role-based access control.
- Production-line configuration data — camera settings, lighting parameters, and line identifiers.
- Quality report data — aggregated shift-wise, daily, and batch-level quality statistics.
- Audit logs — records of classification overrides, configuration changes, and access-control actions.
- ERP/MES synchronization data — production and quality data exchanged with external enterprise systems.

3.8 External Interface Requirements

User Interfaces

- A responsive, web-based quality dashboard for production managers and QC inspectors.
- An administrative configuration panel for camera, lighting, and access-control management.
- Real-time alert notifications delivered via on-screen UI, SMS, or mobile app.

Hardware Interfaces

- Industrial high-resolution cameras mounted along the fabric production line.
- Edge computing devices / IoT gateways for local image pre-processing.

Software Interfaces

- ERP/MES integration via REST/JSON APIs for production and quality data synchronization.
- Cloud storage and compute services for AI model hosting and analytics.

Communication Interfaces

- Secure TLS-encrypted channels for all data transmission between edge, cloud, and client applications.

3.9 System Features

Feature	Summary	Priority
Automated Image Acquisition	Continuous, camera-based fabric image capture synced to line speed.	High
AI Defect Detection	Real-time computer-vision defect identification.	High
Defect Classification & Scoring	Categorization and severity scoring of detected defects.	High
Live Quality Dashboard	Real-time visualization of defect rates and quality trends.	High
Alerting	Threshold-based real-time notifications to operators/managers.	High
Automated Reporting	Scheduled generation and export of quality-inspection reports.	Medium
Access Control	Role-based permissions across all user categories.	High
ERP/MES Integration	API-based synchronization with existing enterprise systems.	Medium
Model Retraining	Ability to retrain the AI model with new labelled data.	Medium

4. Assumptions and Constraints

4.1 Assumptions

- Production lines have adequate space, power, and lighting conditions to support camera and edge-device installation.
- Reliable network connectivity (LAN or cloud) is available at the manufacturing facility.
- A sufficient volume of labelled defect images can be collected to train an initial, reasonably accurate detection model.
- Shop-floor staff will receive basic training on the new dashboard and alerting interfaces before rollout.
- Existing ERP/MES systems expose APIs or middleware capable of integration with the new platform.

4.2 Constraints

Constraint Type	Description
Technical	Detection accuracy depends on camera resolution, lighting consistency, and the diversity of the training dataset; performance may degrade for defect types underrepresented in training data.
Operational	Camera and sensor installation must occur during scheduled downtime to avoid disrupting ongoing production.
Economic	Budget limits may require a phased rollout, starting with a pilot production line before facility-wide deployment.
Legal / Compliance	Cloud-stored production and quality data must comply with applicable data-protection and industry-compliance regulations.
Time	Project timelines constrain the initial release to a single-line pilot rather than a full multi-factory deployment.

5. Requirement Validation

5.1 Completeness Check

Each stakeholder group's core need was cross-checked against the functional requirement set to confirm coverage:

Stakeholder Need	Addressed By
Reduce manual inspection effort	FR-01, FR-02, FR-04 (automated capture, pre-processing, and detection)
Trustworthy, verifiable AI decisions	FR-07 (manual confirm/override), UC-05
Real-time visibility into production quality	FR-08, FR-09 (dashboard and alerts)
Structured data for reporting and audits	FR-10, FR-11 (automated reports and export)
Secure, role-appropriate system access	FR-12 (role-based access control)
Integration with existing enterprise systems	FR-13 (ERP/MES integration)
Sustained model accuracy over time	FR-14 (model retraining)

5.2 Consistency, Ambiguity, and Redundancy Check

- **Consistency:** No conflicting requirements were identified; the real-time detection requirement (FR-04, sub-second latency) and the batch-reporting requirement (FR-10, scheduled generation) operate on independent timeframes and do not contradict one another.
- **Ambiguity:** The original problem statement's phrase "monitor production quality" was decomposed into explicit, testable requirements (FR-08, FR-09) to remove ambiguity about what "monitoring" concretely delivers.
- **Redundancy:** No duplicate requirements were found; each functional requirement maps to a distinct, non-overlapping system capability.

5.3 Suggested Improvements

- Add an explicit requirement defining a minimum acceptable defect-detection accuracy (e.g., a target precision/recall threshold) to make model performance objectively testable.
- Add a formal offline/edge-failover requirement specifying exactly which functions must remain available if cloud connectivity is lost.
- Introduce a non-repudiation/audit-logging requirement for all manual classification overrides, strengthening traceability for compliance audits.

Conclusion

This scenario-based requirement analysis translates the AI-Powered Smart Textile Manufacturing Quality Inspection System's problem statement into a structured, validated Software Requirement Specification. By identifying all relevant actors and use cases, defining comprehensive functional and non-functional requirements, and documenting assumptions, constraints, and validation checks, this analysis establishes a solid, unambiguous foundation for subsequent system design and Agile sprint execution.